



Paper

Quantitative Assessment of Aging Effects on Swallowing Muscle Activity Using Flexible High-density sEMG Array and Deep Learning

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Quantitative Assessment of Aging Effects on Swallowing Muscle Activity Using Flexible High-density sEMG Array and Deep Learning

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Abstract—Aging is closely associated with neuromuscular degeneration in swallowing function. However, noninvasive quantitative methods for systematically characterizing age-related electrophysiological alterations in swallowing muscles remain limited. This study proposes an integrated framework combining a 16-channel flexible wearable surface electromyography (sEMG) sensor array with a Dual-stream Feature Tokenizer (FT)-Transformer model to assess swallowing spatiotemporal coordination and quantify age-related neuromuscular changes. Dynamic sEMG signals were recorded from the submental and infrahyoid muscles of 60 healthy participants (30 young adults and 30 older adults), during 10 swallowing tasks involving different bolus volumes and viscosities. Multidimensional physiological feature analysis and dynamic spatiotemporal energy mapping showed that aging significantly prolonged swallowing duration and onset delay, reduced peak muscle activation, shifted spectral characteristics toward higher frequencies, and resulted in more diffuse and asymmetric muscle coordination patterns. Moreover, the Transformer-based model integrating temporal signals and multidomain physiological features achieved an overall accuracy of 91.29% for age-group classification and 87.03% for bolus viscosity recognition. These findings provide quantitative electrophysiological evidence for understanding aging-induced changes in swallowing patterns and demonstrate the translational value of combining flexible wearable sensing with deep learning for objective swallowing assessment, presbyphagia screening, and precision dietary management.

Index Terms—Aging, Flexible electrode array, Swallowing assessment, Deep learning, Surface electromyography.

I. INTRODUCTION

SWALLOWING is a fundamental physiological process that transports ingested material from the oral cavity through the pharynx and esophagus to the stomach[1]. It is conventionally divided into oral

preparatory, oral propulsive, pharyngeal, and esophageal stages[2], [3]. This process requires precise coordination among cortical and subcortical centers, brainstem networks, cranial nerves, and more than 25 pairs of muscles[4]–[6]. Dysphagia can arise when this complex neuromuscular system is disrupted by stroke, neurodegenerative disease, head and neck tumors, or aging[7]–[9]. Aging is a growing clinical concern because adults aged 60 years and older accounted for 21.1% of China's population in 2023, and meta-analyses indicate that swallowing difficulties affect more than 60% of older adults[10], [11]. Dysphagia is associated with aspiration pneumonia, malnutrition, dehydration, hospital readmission, mortality, and reduced quality of life[12]. Even healthy older adults may exhibit neural degeneration, sarcopenia of swallowing-related muscles, and structural remodeling of the oropharynx and esophagus[13]. Although these changes may not cause clinically evident dysphagia, they reduce swallowing reserve and may emerge under increased physiological demands, creating a need for sensitive assessment of subclinical decline[14], [15]. Videofluoroscopic swallowing study (VFSS) and fiberoptic endoscopic evaluation of swallowing (FEES) provide clinically valuable information but are limited by radiation exposure, invasiveness, portability, or patient tolerance, whereas bedside screening tools are subjective and primarily designed to identify overt dysphagia[16], [17]. Therefore, a non-invasive, objective, quantitative, repeatable, and bedside-compatible method is needed for evaluating early age-related changes in swallowing function.

Surface electromyography (sEMG) provides a non-invasive and radiation-free means of recording swallowing-related muscle activity and is suitable for repeated quantitative assessment[18], [19]. It has been used to detect swallowing events and characterize activation intensity, timing, duration, and coordination under physiological and pathological conditions[20], [21]. However, most studies have focused on clinically evident or disease-related dysphagia, while the subtle

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Institutional Review Board for Human Research Protections of Sixth Affiliated Hospital of Sun Yat-sen University under Application No. 2026ZSLYEC-284.

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neuromuscular changes associated with healthy aging remain insufficiently understood. Although aging can alter the temporal coordination of swallowing muscles and bolus volume and viscosity can independently affect swallowing duration and muscular effort[22], [23], these factors have usually been investigated separately or predominantly in young adults. Consequently, the influence of healthy aging on neuromuscular adaptation to bolus volume and viscosity remains unclear. Conventional swallowing sEMG is constrained by limited spatial coverage. Single or bipolar electrodes can quantify localized activity but provide little information on bilateral symmetry, intermuscular coordination, and dynamic spatial redistribution. High-density sEMG (HD-sEMG) uses electrode arrays to record broader regions and can generate dynamic

energy maps, quantify bilateral activation, and improve identification of disease-related abnormalities[24], [25]. However, many existing arrays are flexible but only minimally stretchable. Their limited mechanical compliance can compromise contact with the curved and deforming neck surface, increasing impedance variation and motion artifacts. Recent flexible and stretchable electronics provide improved skin conformity and signal stability during neck movement[26], [27], yet current studies have mainly emphasized device fabrication, signal quality, event recognition, or disease classification. Their use for quantifying age-related changes in activation intensity, temporal behavior, frequency composition, spatial coordination, and task-dependent adaptation remains limited.

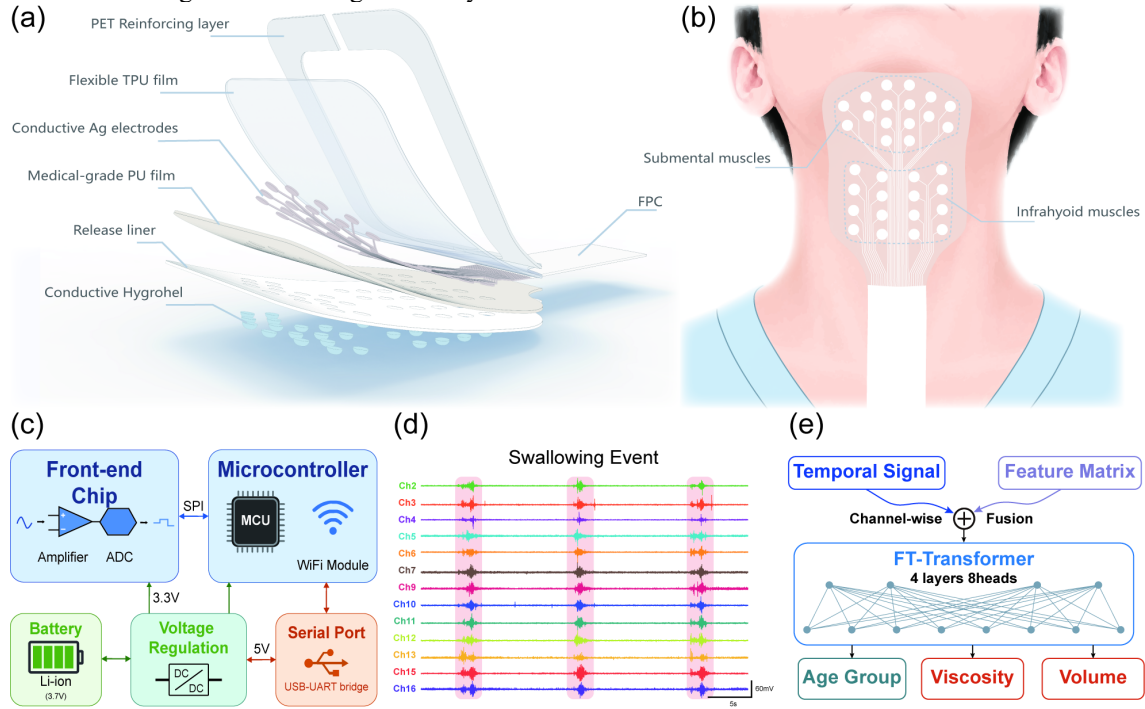


Fig. 1. Overview of the flexible multichannel sEMG sensing and intelligent swallowing assessment system. (a) Structure of the flexible electrode array. (b) Electrode placement over the submental and infrahyoid muscle regions. (c) Architecture of the wireless sEMG acquisition system. (d) Representative channels swallowing sEMG signals. (e) FT-Transformer-based analysis framework for age-group, bolus-viscosity, and bolus-volume classification.

To comprehensively decode the spatiotemporal coordination of swallowing muscles and quantify neuromuscular degeneration associated with aging, this study proposes an integrated framework combining a 16-channel flexible wearable sEMG sensor array with a deep learning model for non-invasive quantitative swallowing assessment (Fig. 1). Leveraging the excellent skin-conformal and anti-motion artifact properties of the flexible array, high-fidelity dynamic sEMG signals were acquired from the submental and infrahyoid muscle groups in 30 healthy young adults and 30 healthy older adults across 10 swallowing tasks involving varying bolus volumes and viscosities. A range of multidimensional physiological features, including swallowing duration, onset delay, amplitude-based muscle work, spectral shifts, and dynamic spatiotemporal energy distributions, were extracted to comprehensively capture electrophysiological alterations induced by aging and bolus property variations. Furthermore, a FT-Transformer model was employed using a dual-stream input strategy that integrates raw temporal signals and multi-domain

physiological features. The proposed model demonstrated high accuracy in age group classification (91.29%) and bolus viscosity recognition (87.03%). Collectively, this study not only advances the understanding of age-related swallowing mechanisms at a physiological level but also highlights the significant translational potential of integrating flexible wearable sensing systems with deep learning for automated screening of age-related dysphagia and precision dietary management in clinical practice.

II. METHODS

A. Participants

A total of 60 healthy participants were recruited from the Sixth Affiliated Hospital of Sun Yat-sen University. The study was approved by the hospital's Ethics Committee (approval No. 2026ZSLYEC-284) and was conducted in accordance with the Declaration of Helsinki. All participants were informed of the study procedures, objectives, and potential risks before the

experiment and provided written informed consent. Participants were stratified by age into a young group and an elderly group. The demographic and anthropometric characteristics are summarized in Table 1. The Young group (n=30): included 13 males and 17 females (age: 21.57 ± 1.36 years; height: 1.66 ± 0.08 m; weight: 60.27 ± 8.97 kg; BMI: $21.40 \pm 1.82 \text{ kg/m}^2$). The Elderly group (n=30): included 11 males and 19 females (age: 68.50 ± 6.97 years; height: 1.63 ± 0.06 m; weight: 58.00 ± 7.18 kg; BMI: $21.81 \pm 1.71 \text{ kg/m}^2$). Detailed inclusion and exclusion criteria are provided in Supplementary Methods S1.

Table 1 General characteristics of the participants

Characteristics	Young(n=30)	Elderly(n=30)	p
Age(years)	21.57 ± 1.36	68.50 ± 6.97	<0.05
Sex(M/F)	13/17	11/19	0.63
Height(m)	1.66 ± 0.08	1.63 ± 0.06	0.09
Weight(kg)	60.27 ± 8.97	58.00 ± 7.18	0.51
BMI(kg/m^2)	21.40 ± 1.82	21.81 ± 1.71	0.15
SSA	18	18	>0.05
WST	I	I	>0.05

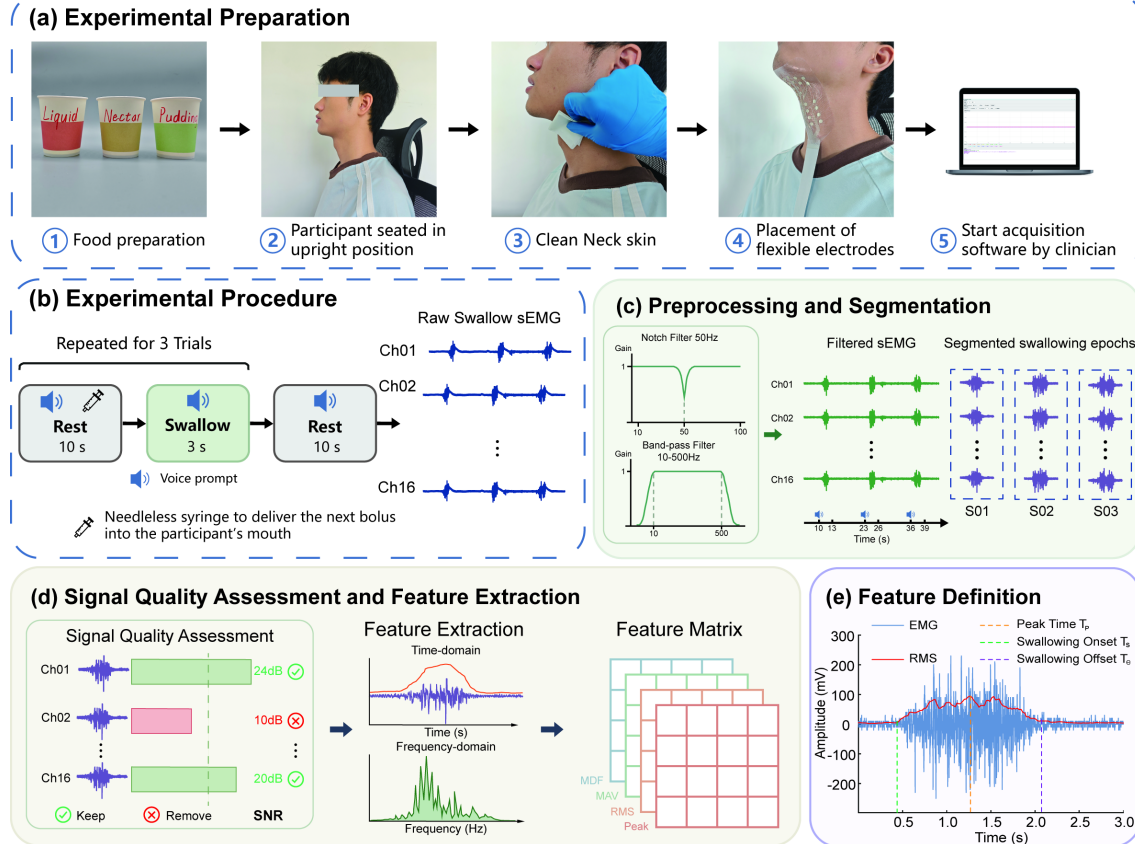


Fig. 2. Experimental protocol and sEMG processing. (a) Experimental preparation. (b) Experimental procedure. (c) sEMG preprocessing and epoch segmentation. (d) Signal quality assessment and feature extraction. (e) Feature definition based on sEMG and RMS signals.

B. Flexible Multichannel sEMG Acquisition System

A custom flexible multichannel sEMG system combined a flexible electrode array for high-fidelity sensing with a wireless module for high-throughput data transmission.

1) Multichannel Flexible Electrode Array

Following our previously optimized fabrication process[26], [28], a neck-conformal array containing 32 contacts was fabricated (Fig. S1). The contacts were 5.4 mm in diameter with 11-mm interelectrode spacing, forming 16 bipolar sEMG channels. Ag electrodes and traces were screen-printed on a 0.025-mm ultrathin TPU substrate and thermally laminated with a biocompatible, self-adhesive, medical-grade PU film (Fig. 1a). Conductive hydrogel was applied to form hydrogel/Ag composite interfaces. A removable 0.07-mm PET perimeter film prevented pre-application deformation and facilitated positioning on the curved neck, to promote electrode conformity and signal stability. The layout was standardized using mean anatomical measurements from 10 participants (Table S1). These included bilateral hyoid extension, bilateral

thyroid cartilage width, and vertical distances between the mandibular and thyroid cartilage boundaries. The 16 bipolar channels were arranged symmetrically over the major swallowing-related muscle regions (Fig. 1b). Channels 1-4 and 13-16 covered the left and right infrahyoid regions, respectively, whereas channels 5-8 and 9-12 covered the left and right submental regions, respectively.

2) Wireless Data Acquisition System

The wireless multichannel sEMG acquisition module integrates signal conditioning, embedded processing, power management, and wireless communication (Fig. 1c). A RHD2216 biopotential front-end chip (Intan Technologies), incorporating 16 low-noise differential channels and a 16-bit analog-to-digital converter, was used to acquire sEMG signals at 1000 Hz via a Serial Peripheral Interface. The on-chip band-pass filter was set to 0.1-750 Hz. An ESP32 microcontroller (Espressif Systems) performed real-time data acquisition, packet processing, and Wi-Fi transmission. The module was powered by a 3.7 V, 600 mAh rechargeable lithium-ion battery, with regulated 3.3 V and 5 V outputs, and included a serial

interface for programming, debugging, and battery management. A custom Python-based graphical user interface enabled real-time multichannel signal visualization and artifact monitoring. Standardized voice prompts, including “start swallowing” and “rest,” were automatically delivered to ensure consistent task timing across participants.

C. Experimental Preparation and Procedure

All experiments were conducted in a quiet, temperature-controlled room. Three bolus viscosities were tested: liquid, nectar, and pudding (Fig. 2a). Nectar and pudding viscosity were prepared by mixing 3 g of edible compound thickener (Easy & Safe) with 200 and 75 mL of purified water, respectively. The liquid consisted of purified water alone. Participants sat upright with a neutral head position. The submental skin was cleaned with 75% alcohol to reduce electrode-skin impedance, and male participants removed facial hair from the electrode placement area. A trained rehabilitation physician localized the hyoid bone by ultrasonography, supported by anatomical landmarks and manual palpation. The flexible array was then placed bilaterally over the submental and infrahyoid muscle regions, with the reference electrode on the wrist. The array was connected to the wireless acquisition module, and signal transmission was verified before recording. The wet swallowing tasks followed a modified Volume-Viscosity Swallow Test, Chinese version (VVST-CV). The complete protocol comprised 10 tasks (Fig. 2b): one additional dry saliva swallow as the baseline and nine wet tasks combining three volumes (3, 5, and 10 mL) with three viscosities. Each task was repeated three times. For each wet trial, the physician administered the bolus using a needle-free syringe, and the participant performed one natural swallow after a standardized voice prompt. Each event was recorded within a 3-s acquisition window, followed by a 10-s rest interval to minimize muscle fatigue. sEMG signals were continuously monitored through the graphical user interface to ensure data quality (Fig. 1d and 2b). Data were automatically saved after each task for offline analysis.

D. Data Processing Procedures

1) Data Preprocessing and Signal Quality Assessment

Raw sEMG signals were preprocessed to suppress power-line interference, baseline drift, and motion artifacts. A 50-Hz notch filter was applied first, followed by a fourth-order zero-phase Butterworth band-pass filter (10-500 Hz) to retain physiological sEMG components and suppress low-frequency artifacts (Fig. 2c). Signals were then segmented into swallowing epochs using the voice-prompt timestamps. Channel-specific signal-to-noise ratios (SNRs) were calculated to assess signal quality and compared between the young and older groups (Fig. 2d). Channels with SNRs below 15 dB or severe persistent artifacts were excluded from feature extraction. SNR was calculated as follows:

$$SNR(dB) = 20 \times \log_{10} \frac{\sum_{k=1}^N V_{signal(k)}^2}{\sum_{k=1}^N V_{noise(k)}^2} \quad (1)$$

$V_{signal(k)}$ and $V_{noise(k)}$ denote the amplitude of the sEMG recorded during active muscle contraction and rest, respectively,

and $N = 100$ denotes the number of samples used for each state.

2) Multidimensional Feature Extraction and Analysis

Time and frequency domain sEMG features were extracted and assembled into a feature matrix to characterize differences in swallowing control between the young and older groups (Fig. 2d). Because segmented epochs could include preparatory or post-swallow rest, active contraction intervals were identified using semi-automated threshold detection followed by expert verification.

Preliminary onset and offset were detected from the root mean square (RMS) envelope (Fig. 2e) using a threshold of the resting mean plus two standard deviations (mean $\pm$ 2SD). Trained investigators blinded to group allocation independently refined these boundaries. Swallowing onset (T_s) was defined as a sharp upward deviation from baseline, and offset (T_e) as the return to a stable baseline. Swallowing duration was calculated as $(T_e - T_s)$ (Fig. 2e). Peak time (T_p) marked maximal muscle activation, whereas swallowing delay was the interval from the voice prompt to (T_s).

In the time domain, features reflecting peak muscular effort were extracted, specifically peak amplitude (the maximum value of the sEMG envelope) and its temporal occurrence (peak time). Additionally, RMS and mean absolute value (MAV) were computed to quantify overall muscle contraction intensity and motor unit recruitment. In the frequency domain, dominant frequency (DF) and median frequency (MDF) were extracted. Both spectral parameters are closely associated with the conduction velocity of motor unit action potentials. The formulas for key feature calculations are given below:

Root mean square (RMS):

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (2)$$

Mean absolute value (MAV):

$$MAV = \frac{1}{N} \sum_{i=1}^N |x_i| \quad (3)$$

x_i is the amplitude of sEMG

Dominant frequency (DF):

$$DF = f_k, \quad k = \arg \max_k PSD(f_k) \quad (4)$$

Median frequency (MDF):

$$\int_0^{MDF} PSD(f) df = \int_{MDF}^{\infty} PSD(f) df = \frac{1}{2} \int_0^{\infty} PSD(f) df \quad (5)$$

$PSD(f)$ is the Power Spectrum of sEMG

To systematically evaluate the extracted sEMG features, non-parametric statistical methods were employed. For between-group comparisons, the Mann-Whitney U test was used to assess neuromuscular differences between the young and elderly groups across specific swallowing tasks. For within-group variations across different bolus volumes and viscosities were evaluated via the Wilcoxon signed-rank test. All statistical tests were two-tailed, with the threshold for statistical significance uniformly set at $p < 0.05$.

E. Deep Learning-based sEMG Classification Model

Dual-stream FT-Transformer architecture[29], [30] was developed for sEMG representation learning and classification. The proposed model employs a dual-stream input strategy that

integrates raw temporal signals with handcrafted physiological features to enhance discriminative feature extraction (Fig. 3). The preprocessed 16-channel sEMG signals were segmented into temporal windows of 100 ms. In the temporal stream, the raw channel-wise signals were embedded into a 128-dimensional latent space using a linear Channel Tokenizer, generating 16 channel tokens. In parallel, 24 multidomain features were extracted from each channel, including six time-domain features, nine FFT-based features, five STFT-based features, and four wavelet features. These features were normalized and projected into the same 128-dimensional embedding space through a feature mapper. The two streams

were subsequently fused by element-wise addition, enabling each channel token to jointly encode temporal dynamics and time-frequency characteristics. A learnable classification token ([CLS]) was appended to the channel token sequence, followed by positional encoding to preserve inter-channel spatial relationships. The resulting sequence was processed by a four-layer Transformer encoder with an eight-head self-attention mechanism to capture complex coordination patterns among swallowing-related muscle channels. Finally, the [CLS] representation was fed into an MLP classification head consisting of layer normalization, GELU activation, dropout ($p = 0.2$), and a linear layer to generate classification outputs.

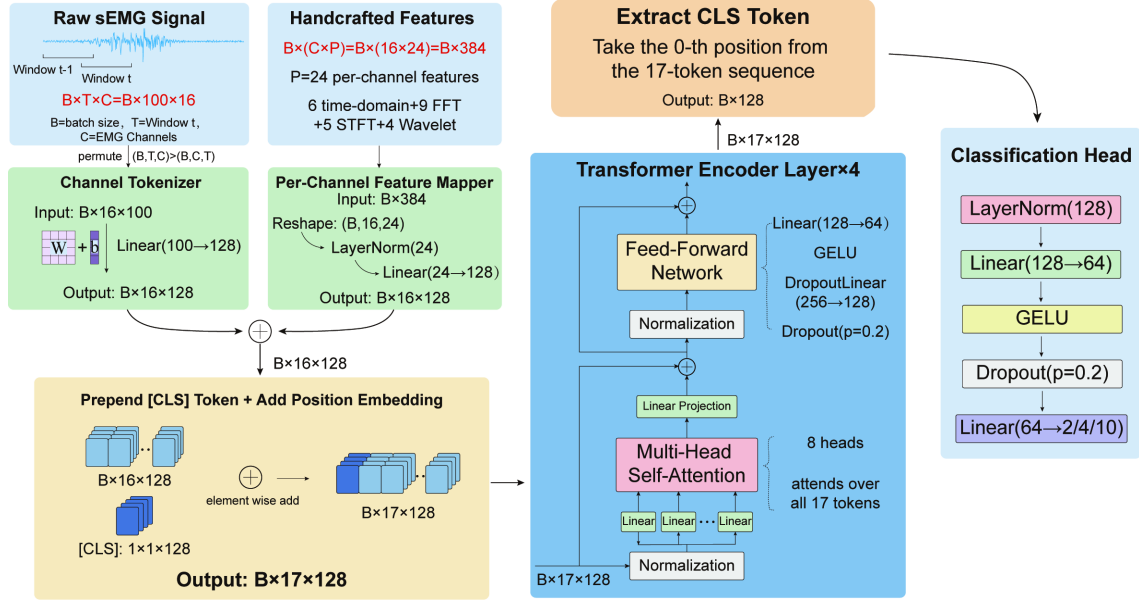


Fig. 3. Architecture of the dual-stream FT-Transformer model for swallowing sEMG classification.

III. RESULTS

A total of 60 participants were recruited, including 30 young and 30 older adults. Signal quality during swallowing was assessed using the mean SNR of each channel (Fig. 4a). SNRs were significantly higher in the young group for most channels ($p < 0.05$), while linear-regression slopes remained close to zero in both groups. Values ranged from 22.89 to 32.39 dB in the young group (mean: 27.52 ± 2.33 dB), with CH3, CH5, CH10, and CH12 exceeding 29 dB. The elderly group had lower values ranging from 19.12 to 28.08 dB (mean: 23.49 ± 2.08 dB).

Swallowing delay was longer in the older group across all dry and bolus-swallowing tasks, with an overall between-group difference of 75.99 ± 25.66 ms (Fig. 4b, c). Dry swallowing produced the longest delays, reaching 896.97 and 927.99 ms in the young and elderly groups, respectively. For bolus swallowing, viscosity affected delay more strongly than volume. From liquid to pudding, mean delay increased from 676.79 ± 9.62 to 768.13 ± 1.29 ms in the young group and from 781.37 ± 25.06 to 830.78 ± 10.05 ms in the elderly group. Increasing volume from 3 to 10 mL within each viscosity produced only small changes without a consistent trend ($p > 0.05$; Fig. S2). Swallowing duration was also generally longer in the elderly group (Fig. 4d-f and Fig. S3). Across tasks, mean duration was 928.16 ms in the young group and 1001.80 ms in the elderly group, an overall difference of 73.64 ± 28.88 ms. The elderly

group also showed greater intertrial variability. Between-group differences were significant for most tasks ($p < 0.05$), except the 3mL and 5mL nectar conditions ($p > 0.05$). Thus, the age-related difference in swallowing duration varied across bolus conditions.

Peak amplitude was significantly higher in the young group across all tasks ($p < 0.05$), with an overall mean difference of 70.84 ± 17.84 mV (Fig. 4g-i and Fig. S4). In the young group, peak amplitude increased from 217.91 ± 8.61 mV for liquid to 250.01 ± 12.74 mV for pudding and from 221.08 ± 14.22 mV at 3 mL to 245.41 ± 18.37 mV at 10 mL. The elderly group showed no significant viscosity-dependent differences ($p > 0.05$) but retained a modest volume-dependent increase from 151.62 ± 7.52 mV at 3 mL to 170.21 ± 2.96 mV at 10 mL. Peak time was generally longer in the elderly group, with a mean between-group delay of 32.06 ms (Fig. 4j, k and Fig. S5). Dry swallowing produced the latest peaks, at 1448.56 and 1468.56 ms in the young and elderly groups, respectively. Across the remaining tasks, peak times ranged from 1201.73 to 1335.39 ms in the young group and from 1282.81 to 1378.79 ms in the elderly group. MAV was consistently higher in the young group across tasks (Fig. 4l and Fig. S6). With increasing bolus volume and viscosity, MAV in the young group increased from 30.46 ± 9.55 mV during dry swallowing to 49.77 ± 12.07 mV during the 10-mL pudding condition. The elderly group showed a narrower range of 23.07 ± 12.61 to 26.28 ± 18.31 mV, indicating limited modulation by bolus properties. Its larger

standard deviations indicated greater interindividual variability despite relatively stable group-level MAV values.

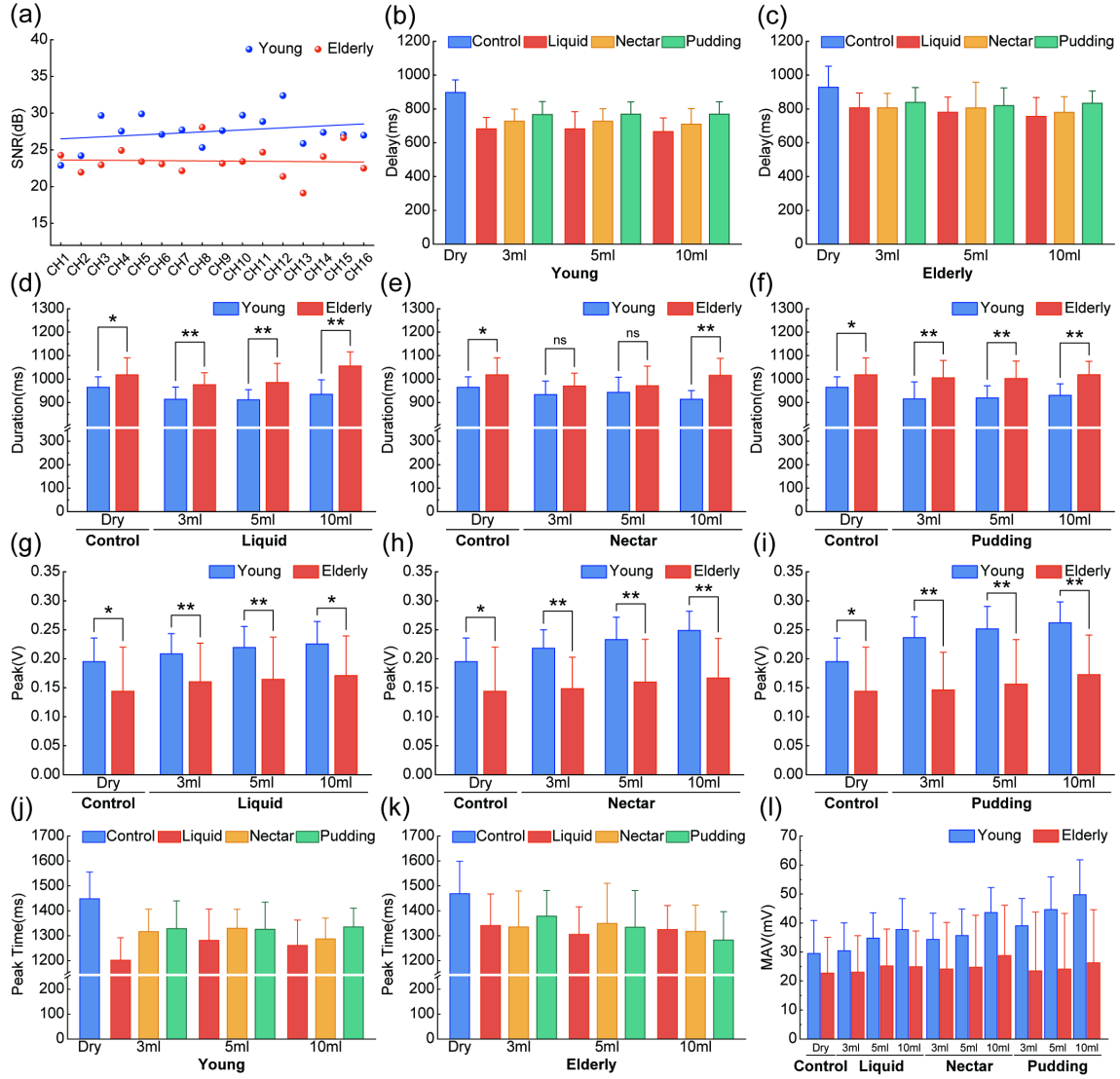


Fig. 4. Time-domain analysis of swallowing sEMG signals. (a) Channel-wise SNR comparison. (b, c) Swallowing delay comparison in young and elderly groups. (d-f) Swallowing duration across different bolus conditions (g-i) Peak amplitude comparison. (j, k) Peak time comparison. (l) MAV comparison between young and elderly groups. Data are presented as mean $\pm$ SD. * $p < 0.05$, ** $p < 0.01$; ns, not significant.

Spectral analyses showed that normalized sEMG power was concentrated at 10-150 Hz, with dominant and median frequencies near 25 and 45 Hz, respectively (Fig. 5a and Fig. S7). Time-frequency spectrograms identified three distinct swallowing events with power concentrated below 150 Hz, whereas inter-swallow resting periods showed low power across 0-500 Hz (Fig. 5b). Dominant frequency was consistently higher in the elderly group across bolus volumes and viscosities (Fig. 5c-e). Values remained relatively stable in the young group, ranging from 19.80 ± 3.39 to 25.09 ± 4.63 Hz, but were higher and more variable in the elderly group, ranging from 24.55 ± 5.77 to 29.70 ± 6.97 Hz. The between-group difference increased with viscosity. Under the pudding condition, mean dominant frequency was 28.93 Hz in the elderly group and 20.28 Hz in the young group. Median frequency was likewise consistently higher in the elderly group (Fig. 5f-h), ranging from 69.73 to 73.11 Hz, compared with

47.70 to 58.57 Hz in the young group. The difference was particularly evident under the pudding condition.

Normalized RMS-based HD-sEMG energy maps revealed distinct spatiotemporal activation patterns between groups (Fig. 6b). In the young group, activation was synchronized, bilaterally symmetric, centrally concentrated, and brief. Activity progressed from the submental to the infrahyoid region, with coordinated peak activation at 0.6-0.9 s. It declined rapidly at 1.2-1.5 s and returned to baseline by 1.8 s. In contrast, the elderly group showed diffuse, irregular, asymmetric, and prolonged activation. Activity remained spatially dispersed and bilaterally uneven during 0.6-1.2 s, with peak energy delayed to approximately 1.2 s. Residual neck activation at 1.5-1.8 s indicated delayed relaxation and incomplete baseline recovery. Using five-fold cross-validation, the dual-stream FT-Transformer distinguished the young and elderly groups with 91.29% accuracy and sensitivities of 91.6% and 90.9%, respectively (Fig. 6c), demonstrating age-discriminative sEMG

features. Classification was more accurate for viscosity than volume. Four-class viscosity classification achieved 87.03% accuracy, with recognition rates above 90% for liquid and pudding (Fig. 6d). Four-class volume classification achieved 63.60% accuracy, with misclassifications occurring primarily among the three volume levels, indicating substantial overlap

in their sEMG representations. (Fig. 6e). Specifically, 18.2% and 14.1% of 3mL trials were misclassified as 5 and 10 mL, respectively. The combined 10-class task achieved 67.54% accuracy (Fig. 6f), with errors mainly occurring among different bolus volumes within the same viscosity. Dry swallowing had the highest class-specific accuracy at 86.1%.

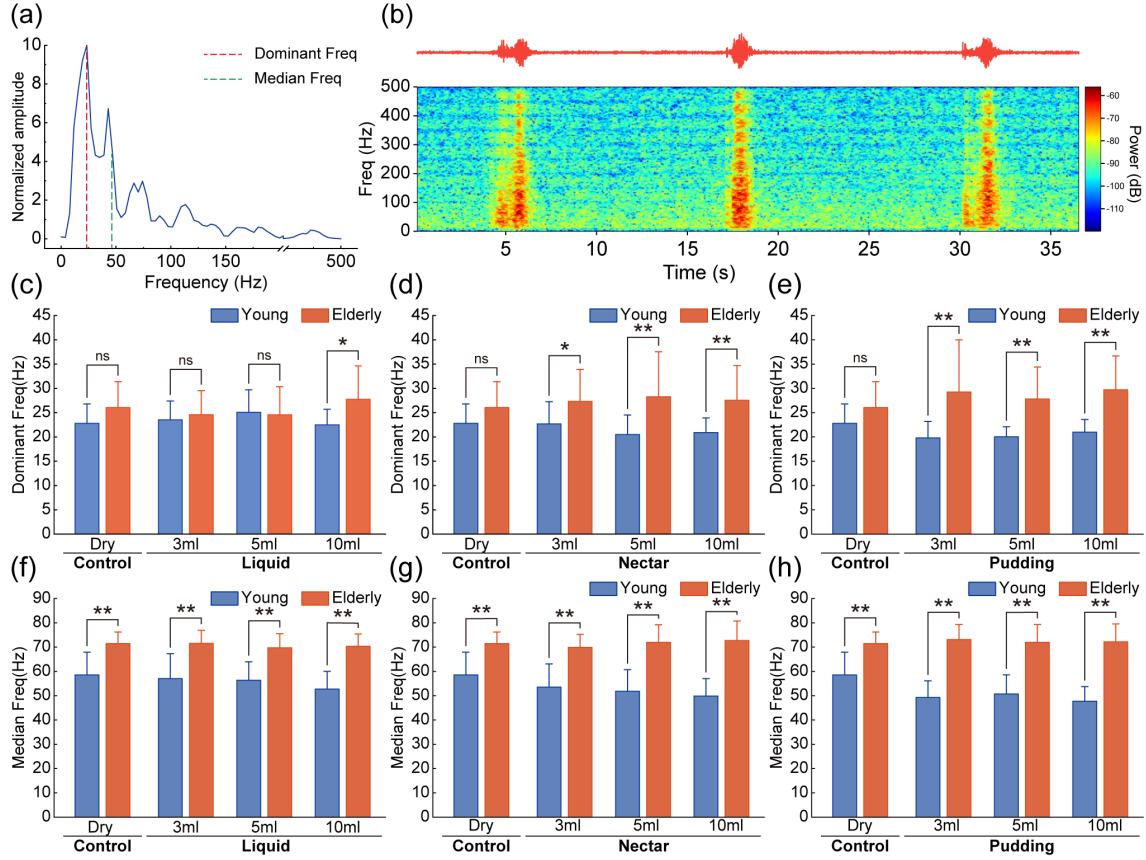


Fig. 5. Frequency-domain analysis of swallowing sEMG signals. (a) Representative normalized power spectrum with dominant frequency and median frequency. (b) Time-frequency representation of repeated swallowing events. (c-e) Dominant frequency comparison. (f-h) Median frequency comparison. Data are presented as mean $\pm$ SD. * $p<0.05$, ** $p<0.01$; ns, not significant.

IV. DISCUSSION

Using a custom 16-channel flexible sEMG system, this study characterized electrophysiological differences in swallowing between the healthy young group and the elderly group. Similar channel-wise signal distributions suggested that the array provided stable recordings during dynamic swallowing. The lower SNR in the elderly group may therefore reflect age-related anatomical and neuromuscular changes rather than poorer measurement performance. Reduced tongue and pharyngeal muscle mass, lower muscle-fiber density, and increased subcutaneous fat can increase source-to-electrode distance and attenuate surface signals[31], [32]. Age-related changes in motor-unit recruitment and firing behavior similarly alter limb-muscle sEMG characteristics [33]. Despite these physiological differences, adequate signal quality in both groups supported reliable neck conformity and acquisition. The array may therefore facilitate swallowing assessment across age groups and characterization of age-related neuromuscular changes.

Age-related changes in swallowing were characterized by slower temporal coordination and reduced adaptability of

neuromuscular activation. Prolonged swallowing delay, duration, and peak time in the elderly group may reflect declines in sensory feedback, sensorimotor integration, and intermuscular coordination[34]. Dry swallowing produced the longest delays and peak times in both groups, likely because it lacks external bolus stimulation and requires saliva accumulation before initiation[24], [35]. Viscosity affected swallowing delay more strongly than volume, suggesting that greater rheological resistance increases demands on oral preparation and neuromuscular coordination[23]. Lower peak amplitudes and MAV values in the elderly group were consistent with reported age-related sEMG reductions. These changes may arise from reduced muscle mass, altered motor-unit recruitment, and lower neuromuscular efficiency[31], [32]. Whereas the young group progressively increased activation with viscosity and volume, the elderly group showed limited modulation, particularly with viscosity. This pattern suggests a reduced capacity to adapt motor output to swallowing demands[34]. The preserved response to volume, however, indicates that viscosity and volume may involve partly distinct sensorimotor mechanisms. Greater MAV variability within the elderly group further indicated heterogeneous age-related changes. This variability may reflect individual compensatory

strategies, including altered motor-unit recruitment and activation timing, that help maintain swallowing function [22]

[33]. Thus, aging affected the magnitude, adaptability, and consistency of swallowing neuromuscular control.

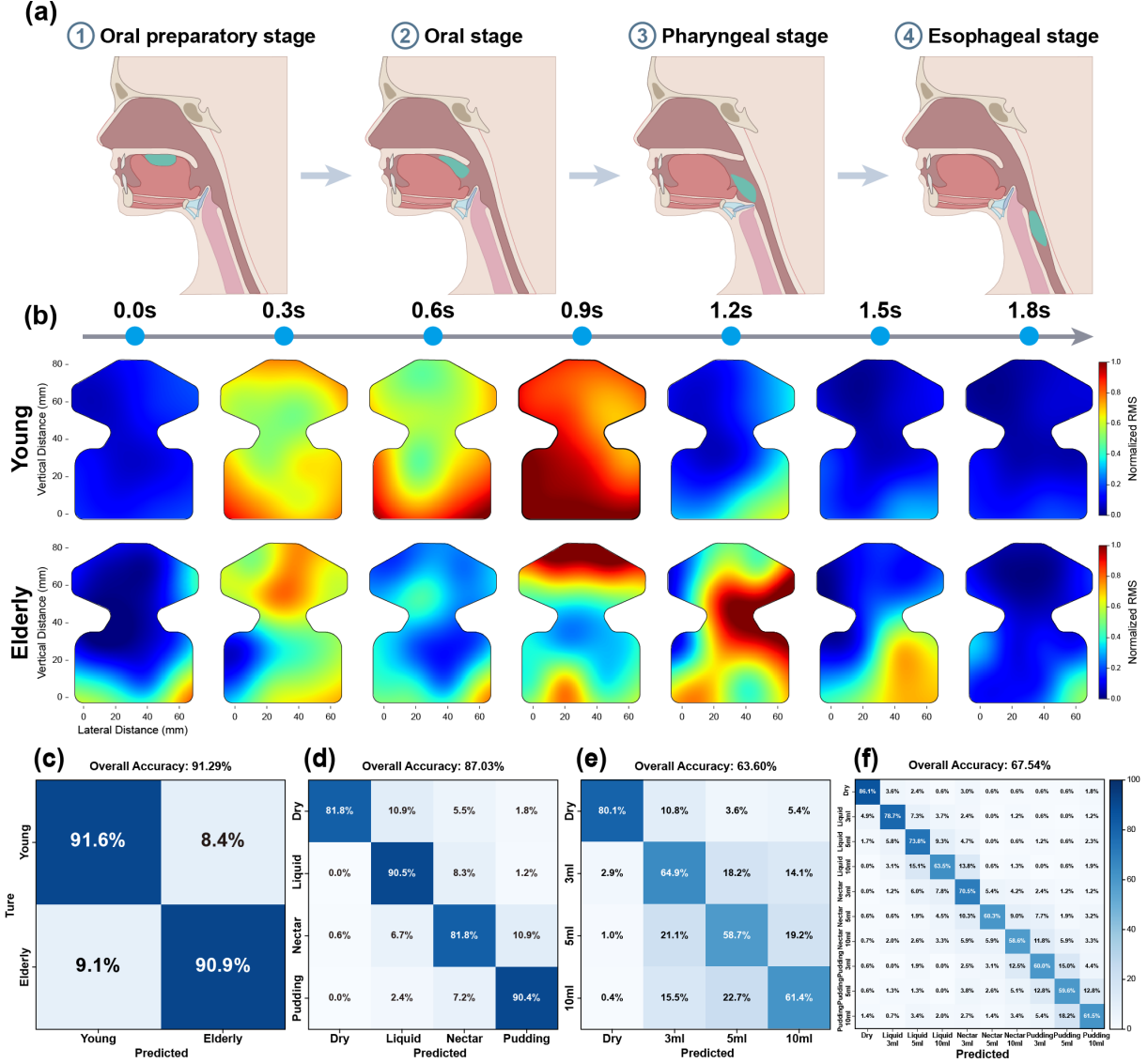


Fig. 6. Dynamic spatiotemporal patterns of swallowing muscle activation and classification performance (a) Physiological stages of swallowing. (b) Dynamic RMS-based sEMG energy maps of swallowing muscle activation in young and elderly groups. (c) Confusion matrix for age-group swallowing task classification. (d) Confusion matrix for viscosity classification. (e) Confusion matrix for volume classification. (f) Confusion matrix for 10-class swallowing task classification.

Swallowing sEMG was concentrated at low frequencies, with transient energy increases during swallowing events. Because spectral features reflect motor-unit recruitment and muscle-fiber conduction [36], the higher dominant and median frequencies in the elderly group may indicate altered motor-unit organization and recruitment. Neuromuscular aging modifies motor-unit firing, recruitment efficiency, and myoelectric characteristics [33], while individual compensatory strategies may contribute to the greater spectral variability observed in the elderly group [31]. These findings underscore the heterogeneous nature of neuromuscular aging. Higher viscosity was associated with wider spectral differences between groups, suggesting distinct neuromuscular adjustments to greater swallowing demands in the elderly group. Dynamic sEMG energy maps complemented the temporal and spectral analyses by visualizing the spatiotemporal organization of muscle activation (Fig. 6). Their evolution corresponded with the

physiological swallowing phases shown in Fig. 6a. In the young group, activation progressed from the submental region during oral preparation and propulsion to a concentrated, symmetric pattern at 0.6–0.9 s during the main propulsive stage. Activity then rapidly returned toward baseline, indicating coordinated activation and relaxation. In contrast, the elderly group showed dispersed, prolonged, and asymmetric activation during 0.6–1.2 s, with delayed peaks and residual post-swallow activity. These patterns were consistent with the longer swallowing duration and peak time, supporting age-related changes in swallowing coordination. Unlike single-channel amplitude measures, dynamic energy maps captured the spatial evolution of activation and provided complementary information for quantitative swallowing assessment.

A dual-stream FT-Transformer integrating dynamic sEMG signals with multidomain features accurately distinguished the young group from the elderly group. This performance

indicated that amplitude, temporal, frequency, and spatiotemporal features captured age-related differences in swallowing. Bolus classification further revealed greater separability by viscosity than volume, with especially high accuracy for liquid and pudding. This finding is consistent with evidence that viscosity produces larger changes in muscle activation by increasing bolus-manipulation and propulsion demands[24]. Lower volume-classification accuracy and frequent errors among volume levels suggest that volume-related changes were more gradual and less separable. The combined 10-class task showed a similar pattern, with most errors occurring between volumes within the same viscosity. Together, these results support multidimensional sEMG features as quantitative descriptors of swallowing variation. Integrating high-density sEMG with deep learning may therefore facilitate objective assessment across age groups and task conditions.

V. CONCLUSION

This study validated the efficacy of integrating a customized 16-channel flexible wearable sEMG system with an FT-Transformer model for the quantitative assessment of age-related swallowing function. Benefiting from the ultrathin and skin-conformal flexible electrode array, robust multichannel sEMG signal acquisition was achieved during dynamic swallowing tasks, enabling a systematic characterization of electrophysiological differences in swallowing muscles between young and older adults. The results demonstrated that aging was associated with reduced muscle activation intensity, prolonged swallowing duration, altered spectral characteristics, and changes in spatiotemporal coordination patterns. Furthermore, the FT-Transformer-based deep learning architecture effectively integrated raw sEMG time-series signals with multidimensional electrophysiological features, achieving accuracies of 91.29% and 87.03% in age-group classification and bolus viscosity recognition, respectively. These results further confirm that aging-related information embedded in swallowing sEMG signals is highly separable and recognizable. Collectively, these findings provide quantitative insights into the physiological mechanisms underlying aging-induced alterations in swallowing patterns and establish a potentially valuable technical framework for the clinical assessment of presbyphagia. Future studies will further extend the application of this system to clinical populations with dysphagia, particularly patients with specific neurological disorders such as post-stroke dysphagia. In addition, subsequent work will enlarge the sample size, incorporate subject-specific anatomical information, and optimize the real-time analytical capability of the system to support personalized swallowing rehabilitation assessment and precision dietary intervention.

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Quantitative Assessment of Aging Effects on Swallowing Muscle Activity Using Flexible High-density sEMG Array and Deep Learning

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SUPPLEMENTARY METHODS S1

Inclusion criteria: (1) Healthy young adults aged 18-25 years or healthy elderly adults aged ≥ 60 years; (2) Normal swallowing function, confirmed by regular dietary intake without a history of choking or dysphagia; (3) A score of 18 on the Standardized Swallowing Assessment (SSA) and a Grade I on the Water Swallow Test (WST); (4) Intact cognitive and communication abilities sufficient to understand experimental instructions; (5) The ability to maintain an independent seated posture for at least 30 min.

Exclusion criteria: (1) Neurological disorders affecting swallowing or motor control (e.g., stroke, Parkinson's disease, and dementia); (2) Previous head and neck surgery, trauma, or radiotherapy; (3) Acute upper respiratory infection within the preceding week or active symptoms (e.g., pharyngeal pain) that could influence with swallowing.

Participant	A	B	C	D
1	5.5	5.5	3.5	5.5
2	6.0	6.5	3.0	6.5
3	5.5	6.5	3.5	6.0
4	5.0	5.5	3.0	6.5
5	7.5	6.5	5.0	7.0
6	5.5	8.0	3.3	7.0
7	6.5	7.5	4.0	7.0
8	6.0	6.0	3.5	7.0
9	7.0	7.0	3.5	5.0
10	7.0	7.0	4.0	6.0
Mean	6.15	6.60	3.63	6.30

A: The bilateral hyoid extension(cm), B: The bilateral thyroid cartilage width(cm), C: The vertical distances between the mandible(cm), D: The thyroid cartilage boundaries(cm)

Table S1 Anatomical measurements obtained from 10 participants for standardizing the electrode array layout.

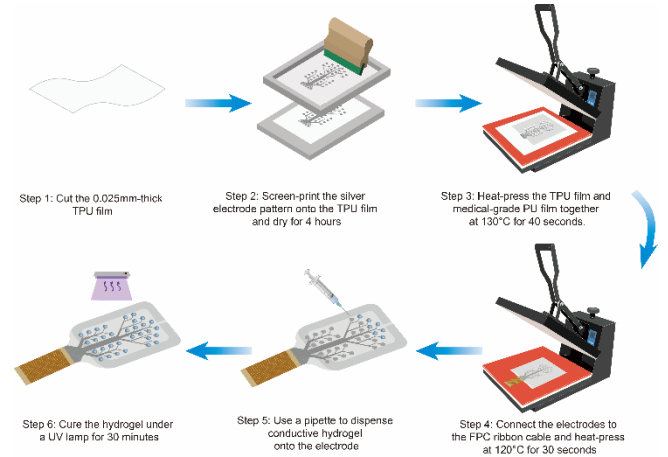


Fig. S1. Fabrication process of the flexible multichannel sEMG electrode array.

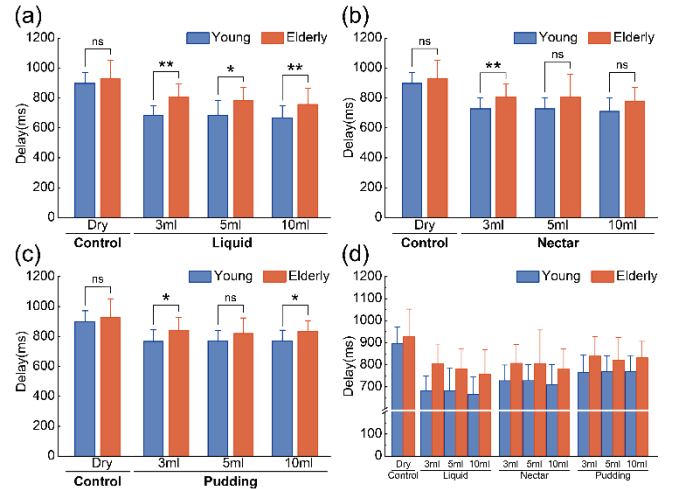


Fig. S2. Swallowing delay in young and elderly participants under different bolus conditions. (a-c) Between-group comparisons during dry swallowing and swallowing of (a) liquid, (b) nectar viscosity, and (c) pudding viscosity at 3, 5, and 10 mL. (d) Summary of swallowing delay across all tasks. Data are presented as mean $\pm$ SD. * $p < 0.05$, ** $p < 0.01$; ns, not significant.

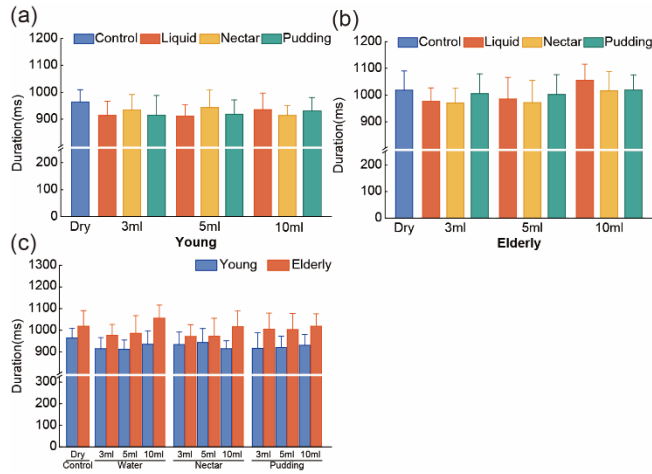


Fig. S3. Swallowing duration under different bolus conditions in young and elderly participants. (a, b) Swallowing duration during dry swallowing and liquid, nectar-thick, and pudding-thick bolus swallowing at 3, 5, and 10 mL in the young and elderly groups, respectively. (c) Between-group comparison across all swallowing tasks. Data are presented as mean $\pm$ SD.

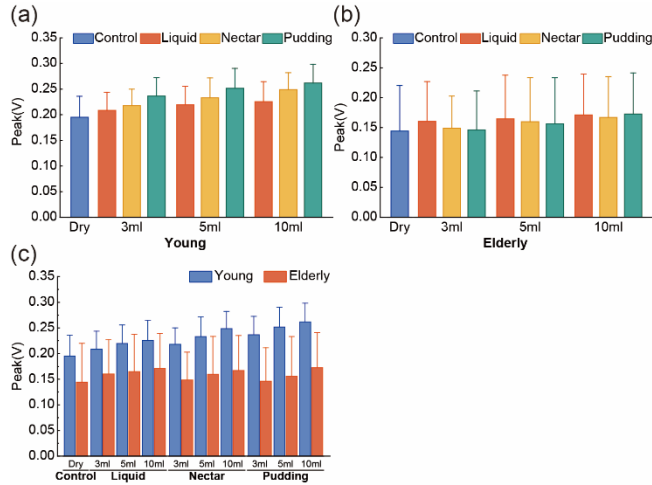


Fig. S4. Peak amplitude of swallowing sEMG signals under different bolus conditions. (a, b) Peak amplitude during dry swallowing and swallowing of liquid, nectar-thick, and pudding-thick boluses at 3, 5, and 10 mL in the young and elderly groups, respectively. (c) Between-group comparison across all swallowing tasks. Data are presented as mean $\pm$ SD.

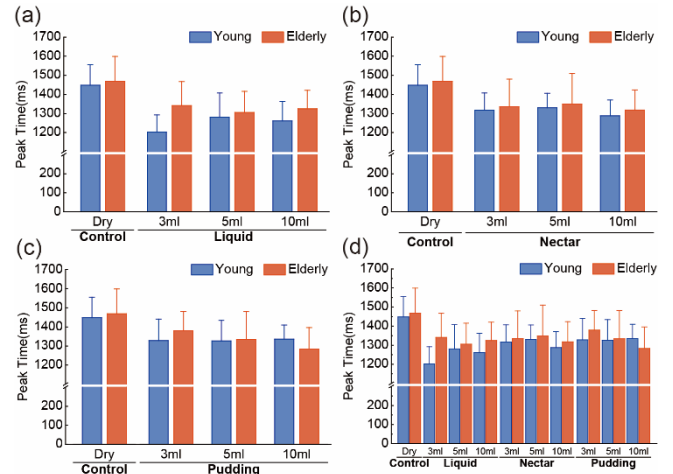


Fig. S5. Peak time of swallowing sEMG signals in young and elderly participants. (a-c) Peak time during dry swallowing and swallowing of (a) liquid, (b) nectar viscosity, and (c) pudding viscosity at 3, 5, and 10 mL. (d) Comparison across all swallowing tasks. Data are presented as mean $\pm$ SD.

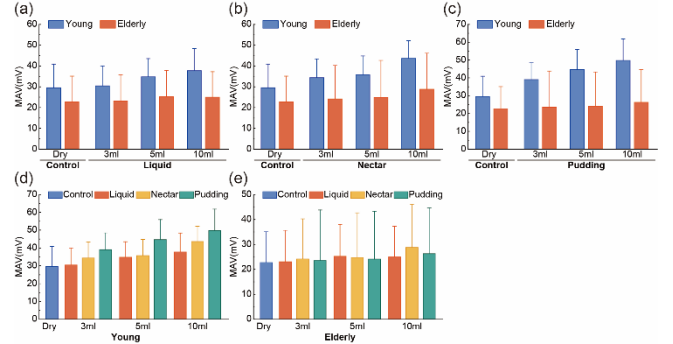


Fig. S6. Mean absolute value (MAV) of swallowing sEMG signals under different bolus conditions. (a-c) Comparison between young and elderly participants for (a) liquid, (b) nectar viscosity, and (c) pudding viscosity. (d, e) MAV across all swallowing tasks in the young and elderly groups, respectively. Data are presented as mean $\pm$ SD.

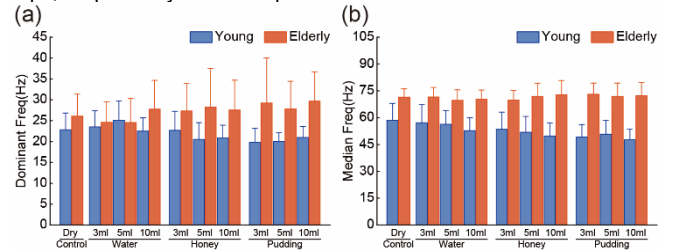


Fig. S7. Frequency-domain characteristics of swallowing sEMG signals in young and elderly participants. (a) Dominant frequency and (b) median frequency during dry swallowing and swallowing of water, honey-thick, and pudding-thick boluses at 3, 5, and 10 mL. Data are presented as mean $\pm$ SD.

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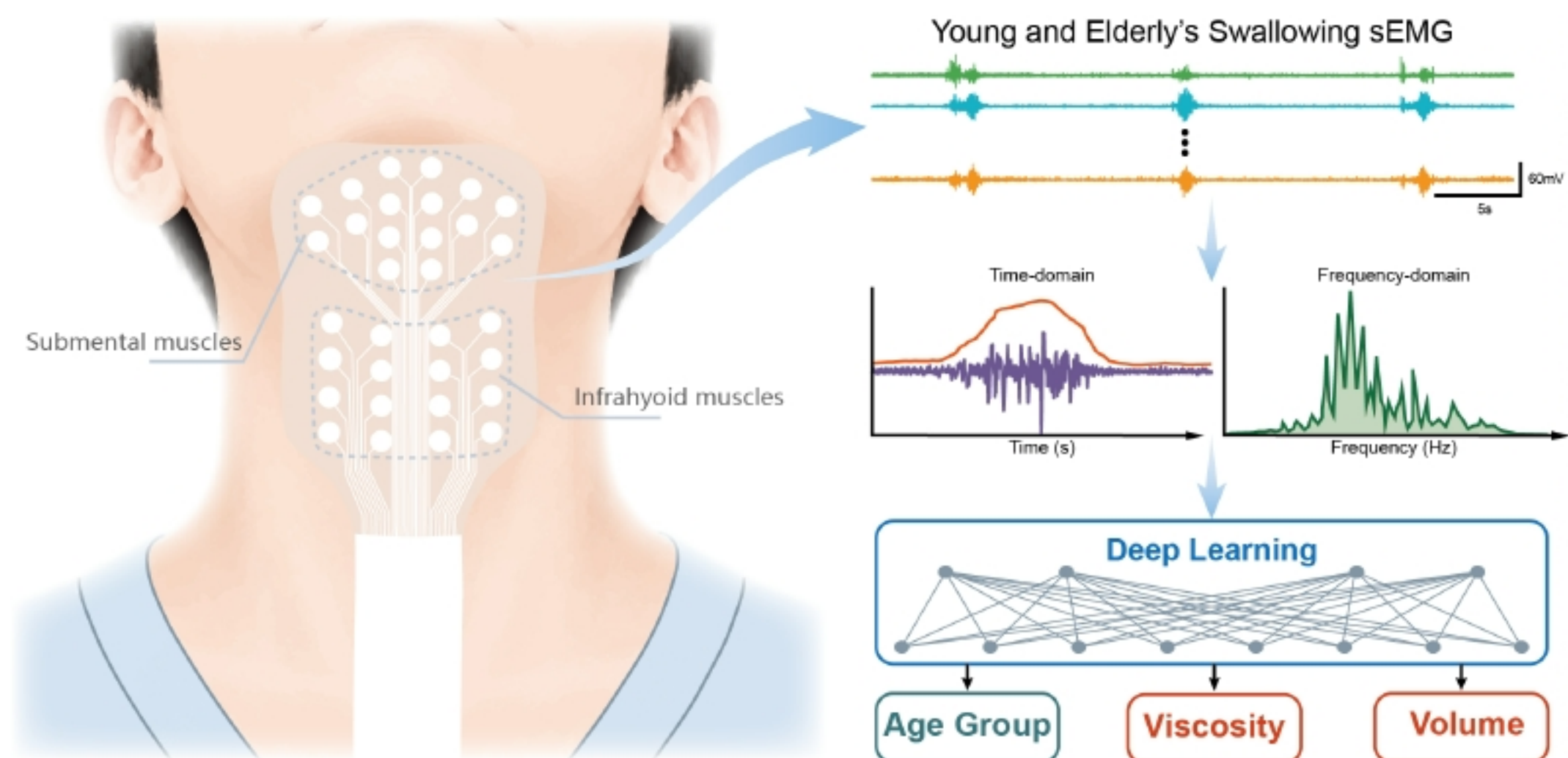
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A flexible high-density sEMG array records swallowing activity from the submental and infrahyoid muscles in young and elderly. Time and frequency domain features extracted from these signals are integrated through deep learning to characterize age-related neuromuscular differences and distinguish age group, bolus viscosity, and volume, enabling noninvasive and quantitative assessment of swallowing function across different physiological and dietary conditions.